WEIGHTED HISTORICAL PERFORMANCE

A quantitative framework for NBA, NFL and NCAAF player-stat forecasting

Strategy 01 | Mathematical documentation and implementation guide

Scope: all player performance props; weighted recency, opportunity adjustments, predictive distributions, calibration, walk-forward backtests and a reusable Python engine.

Prepared October 2026. All numerical player examples are hypothetical and are not live betting recommendations.

1. Executive summary and workflow

Weighted Historical Performance (WHP) gives recent player performances greater importance than older ones. The model estimates a next-game expectation from a normalized weighted history, then extends it with workload projections, shrinkage, and uncertainty modeling. It is a baseline against which more complex player-prop models should be tested.

$$\hat{X} = \Sigma_i w_i X_i,\ w_i \geq 0,\ \Sigma_i w_i = 1$$

Symbol	Meaning
X_i	Statistic from historical game i
w_i	Normalized historical weight
N	Number of historical games
$\hat{X}$	Forecast statistic in next game

Pipeline: collect historical observations → calculate recency weights → estimate rate and opportunity → quantify uncertainty → generate predictive distribution → calculate prop probabilities → validate forecasts.

2. Weighted historical estimators

2.1 Simple moving average

$$\hat{X}_{SMA} = (1/N)\ \Sigma_i X_i$$

A simple average weights every observation equally. For five scoring totals 15, 18, 20, 27, 30, the SMA is 22.000000. This can lag behind a recent role change.

2.2 Custom weighted moving average

$$\hat{X}_{WMA} = \Sigma_i w_i X_i$$

Game age	Points	Weight	Weighted contribution
4	15	5%	0.75
3	18	10%	1.80
2	20	15%	3.00
1	27	25%	6.75
0 (latest)	30	45%	13.50
Total		100%	25.80

Weighted projection = 25.800000, versus an unweighted average of 22.000000. This is a forecast assumption, not proof the recent trend persists.

2.3 Linear weights

$$w_i = 2i / [N(N + 1)],\ i = 1 \text{ oldest}, \ldots, N \text{ newest}$$

Simple and transparent, but the abrupt window cutoff causes old observations to lose all influence.

2.4 Exponential weights and half-life

$$w_i = \exp(-\lambda a_i) / \Sigma_j \exp(-\lambda a_j)$$

$$\lambda = \ln(2)/h,\ w_i = 2^{\wedge}(-a_i/h) / \Sigma_j\ 2^{\wedge}(-a_j/h)$$

Here a_i is the number of appearances since game i, with 0 indicating the most recent game; h is the half-life measured in games. Smaller h adapts quickly, while larger h smooths noise. Potential candidate half-lives: NBA 3/5/8/12; NFL 2/3/5/8; NCAAF 2/3/4/6. These are search-grid starting points, not validated optimal values.

2.5 Calendar-based decay

$$w_i \propto \exp[-\lambda d(T - t_i)]$$

Use days elapsed rather than games elapsed when long gaps affect information relevance. Injury returns and offseason changes often require a separate workload model regardless of weighting method.

3. Dispersion, effective sample size and regularization

$$\hat{\mu} = \Sigma_i\ w_i\ X_i;\ v_w = \Sigma_i\ w_i\ (X_i - \hat{\mu})^2$$

$$s_w^2 = v_w / (1 - \Sigma_i\ w_i^2)$$

The second expression is a finite-sample correction under independent identically distributed assumptions. Sports observations need not satisfy those assumptions; calibrate next-game predictive dispersion on held-out data. Do not confuse standard error of the mean with uncertainty in the actual next-game outcome.

$$N_eff = 1 / \Sigma_i\ w_i^2$$

For weights (0.05, 0.10, 0.15, 0.25, 0.45), N_eff = 3.333333 despite having five observed games. A highly concentrated weighting scheme effectively relies on relatively few results.

3.1 Shrinkage toward a prior

$$\alpha = N_eff / (N_eff + k)$$

$$\hat{\mu}_adj = \alpha\ \hat{\mu}_WHP + (1 - \alpha)\ \mu_0$$

μ_0 may be a prior season or position-role estimate, and k controls prior strength. With WHP = 25.8, μ_0 = 20, N_eff = 3.333333 and k = 5: α = 0.4 and $\hat{\mu}$_adj = 22.320000. Use a prior constructed only from information available before prediction time.

4. NBA model

Suitable markets: points, rebounds, assists, threes, steals, blocks, turnovers, and combinations. Direct WHP predicts each statistic X as $\Sigma w_i X_i$. Minutes-adjusted modeling improves interpretability when rotation changes.

$$\hat{r} = (\Sigma_i\, w_i\, X_i) / (\Sigma_i\, w_i\, M_i);\ \hat{X}_rate = \hat{r} \times \hat{M}$$

M_i = historical minutes, $\hat{M}$ = separately forecast minutes. This ratio-of-totals avoids overemphasizing short stints with abnormally high per-minute rates. A scoring rate of 0.72 points/minute with $\hat{M} = 34$ produces 24.480000 points.

Market	Opportunity	Efficiency
Points	Minutes / shooting chances	Points per minute or shooting opportunity
Rebounds	Minutes / rebound chances	Rebound conversion
Assists	Minutes / potential assists	Assist conversion
Three-pointers	3PT attempts	3PT make rate
Steals / blocks	Defensive possessions / shots	Events per opportunity
Turnovers	Touches / possessions	Turnover rate

$$\hat{X}_NBA = \beta\, \hat{X}_direct + (1 - \beta)\, \hat{X}_rate,\ 0 \le \beta \le 1$$

Fit β out of sample. Adjust projected minutes for expected starts, injuries, restrictions and rotation changes, using only information available at forecast time.

5. NFL model

Model opportunity separately from efficiency when future attempts or targets differ from historical levels. Under identical game samples and weights, multiplying the weighted historical opportunity mean by a weighted aggregate efficiency ratio often simplifies back to the direct historical weighted stat mean; the decomposition adds value when opportunities are separately forecast.

5.1 Quarterback passing yards

$$\hat{Y}_pass = \hat{A}_pass \times \widehat{YPA}$$

$$\widehat{YPA} = (\Sigma_i w_i Y_pass,i) / (\Sigma_i w_i A_pass,i)$$

Example: 37 projected passing attempts × 7.2 yards/attempt = 266.400000 yards. Forecast completions similarly via attempts × completion rate.

5.2 Running back rushing yards

$$\hat{Y}_rush = \hat{A}_rush \times \widehat{YPC}$$

$$\widehat{YPC} = (\Sigma_i w_i Y_rush,i) / (\Sigma_i w_i A_rush,i)$$

5.3 Wide receiver receiving yards

$$\hat{Y}_{rec} = \hat{T} \times \widehat{CR} \times \widehat{YPR}$$

$$\widehat{CR} = \Sigma w_i C_i / \Sigma w_i T_i;\ \widehat{YPR} = \Sigma w_i Y_i / \Sigma w_i C_i$$

Nine projected targets × 0.68 catch rate × 12.5 yards per reception = 76.500000 yards. This product is a plug-in approximation: the expectation of a product is not generally the product of its component expectations.

Market	Model components
Passing yards	Attempts × yards per attempt
Completions	Attempts × completion rate
Passing TD	Attempts × TD rate; count distribution
Rushing yards	Carries × yards per carry
Receptions	Targets × catch rate
Receiving yards	Targets × catch rate × yards/reception
Anytime TD	Scoring opportunities → probability of at least one
Longest reception	Reception-level simulation; maximum statistic

6. NCAAF model

The NFL decomposition carries over, but college football has greater cross-opponent variation, transfers, rotating quarterbacks and frequent game-script-driven workload changes. Use stronger regularization where historical samples are sparse.

$$\hat{s}_{carry} = \Sigma_i\ w_i\ PlayerCarries_i\ /\ \Sigma_i\ w_i\ TeamCarries_i$$

$$\hat{A}_{rush} = \hat{s}_{carry} \times TeamCarries\hat{s};\ \hat{Y}_{rush} = \hat{A}_{rush} \times YP\hat{C}$$

Historical production against weak opponents may translate poorly against strong ones. Maintain separate opponent and expected play-volume modules as subsequent enhancements.

7. Predictive distributions and Over/Under probabilities

A mean alone cannot estimate the chance of clearing a prop line. Fit a distribution of next-game outcomes using the point projection and calibrated predictive uncertainty.

Market	Candidate distributions / considerations
NBA points	Empirical, Normal approximation or overdispersed count
Rebounds / assists	Poisson or Negative Binomial
3PT makes	Binomial or Beta-Binomial conditioned on attempts

Market	Candidate distributions / considerations
Steals / blocks	Low-count or zero-heavy models
Passing yards	Empirical, mixture or Normal approximation
Rushing yards	Empirical/mixture; net yards may be negative
Receiving yards	Zero-inflated / empirical mixture
Touchdowns	Count model or opportunity-based Bernoulli model

7.1 Normal approximation

$$X_{next} \sim \text{Normal}(\hat{\mu}, \sigma^2_{pred})$$

$$P(X > L) = 1 - \Phi[(L - \hat{\mu}) / \sigma_{pred}]$$

For integer statistics and a half-point line, continuity correction: $P(X \geq k) \approx 1 - \Phi[(k - 0.5 - \hat{\mu})/\sigma_{pred}]$. Use a calibrated next-game σ_{pred}, NOT merely uncertainty in the estimated mean.

7.2 Poisson and Negative Binomial

$$\text{Poisson: } P(X = k) = \exp(-\mu)\, \mu^k / k!, \; \text{Var}(X) = \mu$$

$$\text{Negative Binomial: } \text{Var}(X) = \mu + \mu^2/r; \; \hat{r} = \hat{\mu}^2/(\hat{v} - \hat{\mu})$$

The method-of-moments Negative Binomial estimate requires $\hat{v} > \hat{\mu}$. Select the distribution using out-of-sample calibration and support constraints.

7.3 Weighted nonparametric model

$$\hat{F}_w(x) = \Sigma_i \, w_i \, 1(X_i \leq x)$$

$$\hat{P}(X > L) = \Sigma_i \, w_i \, 1(X_i > L)$$

The empirical model is transparent but cannot naturally project results beyond observed historical outcomes; consider weighted bootstrap or fitted tails for distant alternative lines.

8. Fully worked NBA scoring example

Game age	Points	Weight	Contribution
9	17	2.86%	0.487
8	22	3.61%	0.794
7	19	4.54%	0.863
6	25	5.73%	1.431
5	21	7.21%	1.515
4	28	9.09%	2.545
3	23	11.45%	2.634
2	31	14.43%	4.473
1	26	18.18%	4.726
0	29	22.90%	6.642
Total		100%	26.108708

Unweighted average = 24.100000; weighted average = 26.108708; weighted descriptive SD = 3.746163; effective sample size = 7.125124.

Illustratively, suppose a separately calibrated predictive SD of six points, so $X_{next} \sim Normal(\hat{\mu}, 6^2)$. The estimated probability of Over 24.5 is $1 - \Phi((24.5 - \hat{\mu})/6)$. The same fitted distribution can price multiple alternative thresholds. This is a hypothetical example, not a live projection.

9. Walk-forward calibration and validation

Split historical data chronologically. At date t, train only on observations and ancillary news available strictly before t; issue a prediction for t, log it, reveal the actual result, and advance the clock. Tune hyperparameters on earlier folds and preserve a final untouched evaluation interval.

$$\theta^* = \text{argmin}_\theta \, L_{\text{out-of-sample}}(\theta)$$

Parameters include window N, half-life h, shrinkage k, blending β and dispersion/distribution choices. Tune separately by league and relevant market; football may benefit from position-level pooling.

$$MAE = (1/T) \Sigma_t |X_t - \hat{X}_t|$$

$$RMSE = \text{sqrt}[(1/T) \Sigma_t (X_t - \hat{X}_t)^2]$$

$$Bias = (1/T) \Sigma_t (\hat{X}_t - X_t)$$

$$Brier = (1/T) \Sigma_t (p_t - y_t)^2$$

Assess calibration curves, Brier score, log loss and full-distribution scoring where possible. Whole-number betting lines require separate push handling. Avoid leakage from revised lineup information, final play counts,

late injury reports and final box scores.

10. Market comparison

$$\text{Positive odds } +A\text{: } q = 100/(A + 100)$$

$$\text{Negative odds } -A\text{: } q = A/(A + 100)$$

$$\text{Two-sided proportional no-vig: } q_{\text{fair,over}} = q_{\text{over}}/(q_{\text{over}} + q_{\text{under}})$$

$$\text{Binary no-push EV per \$1 staked} = p\,b - (1 - p)$$

At hypothetical p = 0.60 and odds −110 (b = 100/110), EV = 0.145455 per dollar, conditional on accurate probabilities. Miscalibration, stale lines and model uncertainty can eliminate an apparent advantage.

11. Combined props, covariance and model limits

$$E[PRA] = E[P] + E[R] + E[A]$$

$$\text{Var(PRA)} = \text{Var(P)}+\text{Var(R)}+\text{Var(A)}+2\text{Cov(P,R)}+2\text{Cov(P,A)}+2\text{Cov(R,A)}$$

Linearity of expected value holds regardless of dependence. The full variance requires covariance. Model multi-stat props jointly when possible, preserving game/pace/minutes relationships. Major limitations include outliers, injuries, role changes, sample sparsity, defense quality, correlated stats and tail misspecification.

12. Proposed application and data architecture

Modules: historical_data.py, weights.py, core.py, opportunity.py, distributions.py, calibration.py, backtesting.py, prediction_service.py, and dashboard player_analytics.py. Separate data collection, feature engineering, forecast generation, distribution fitting and market evaluation.

Input field	Purpose
League, PlayerID, TeamID, GameID	Unique dimensions
GameDate, OpponentID, Market	Temporal and matchup context
ActualValue, Opportunities	Historical target and minutes/attempts/targets
ParticipationStatus	Role and appearance data
PredictionTimestamp	Audit the forecast as of a specific time
DataAvailabilityTimestamp	Prevent future-information leakage

Persist outputs: model version, input cutoff, market, weighted mean, adjusted mean, effective sample size, predicted opportunity, calibrated dispersion, line probabilities, and performance diagnostics.

13. Reusable Python implementation

```
import numpy as np
from scipy.stats import norm

class WeightedHistoricalPerformance:
    def __init__(self, half_life=5, window=20, prior_strength=0):
        if half_life <= 0 or window < 1 or prior_strength < 0:
            raise ValueError("Invalid WHP parameters")
        self.half_life = half_life
        self.window = window
        self.prior_strength = prior_strength

    def weights(self, n):
        age = np.arange(n - 1, -1, -1)
        raw = 2.0 ** (-age / self.half_life)
        return raw / raw.sum()

    def predict(self, history, prior=None):
        x = np.asarray(history, dtype=float)[-self.window:]
        if not len(x) or not np.all(np.isfinite(x)):
            raise ValueError("History must be nonempty and finite")
        w = self.weights(len(x))
        mean = np.sum(w*x)
        var = np.sum(w*(x-mean)**2)
        neff = 1.0 / np.sum(w*w)
        adjusted = mean
        if prior is not None:
            alpha = neff / (neff + self.prior_strength)
            adjusted = alpha*mean + (1-alpha)*prior
        return dict(weighted_mean=float(mean), adjusted_mean=float(adjusted),
                    weighted_variance=float(var), descriptive_std=float(np.sqrt(var)),
                    effective_n=float(neff), weights=w)

    @staticmethod
    def over_probability(mean, calibrated_predictive_std, line):
        if calibrated_predictive_std <= 0:
            raise ValueError("Predictive SD must be positive")
        return float(norm.sf(line, loc=mean, scale=calibrated_predictive_std))

model = WeightedHistoricalPerformance(half_life=3, window=10, prior_strength=5)
```

```
result = model.predict([17,22,19,25,21,28,23,31,26,29], prior=22)
print(result)
print(model.over_probability(result['adjusted_mean'], 6.0, 24.5))
```

14. Roadmap and final recommendation

Version	Deliverables
V1: baseline	Exponential weights; window; mean/variance; effective sample size
V2: opportunity	NBA minutes; football attempts, targets and shares; shrinkage
V3: probability	Market-appropriate distributions; predictive variance; alternative lines
V4: production	Walk-forward tests; calibration; model comparison; versioned predictions

$$\bar{X}_{final} = \alpha[\beta\,\bar{X}_{WHP} + (1-\beta)\,\bar{X}_{opportunity}] + (1-\alpha)\mu_0$$

Recommended starting candidate: half-life exponential weighting + opportunity adjustments + historical shrinkage, followed by an independently calibrated predictive distribution. Retain both the unadjusted WHP benchmark and the enhanced forecast to measure whether complexity genuinely improves out-of-sample predictions.

Central design principle: The objective is not merely to forecast an average statistic but to estimate an empirically calibrated next-game outcome distribution for every supported player-prop market.